

Commerce XI Maths App

Theory (Pages 8-15)

Detailed Answers with Explanations and Steps

Mathematics & Statistics, Part 2, Unit 1 - Partition Values | Total questions: 42

Question 1 - Theory P8

Deciles divide ordered observations into

Options

- A. 10 equal parts
- B. 4 equal parts
- C. 2 equal parts
- D. 100 equal parts

Correct answer: 10 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 10 equal parts.

Question 2 - Theory P8

The number of deciles is

Options

- A. 9
- B. 10
- C. 4
- D. 99

Correct answer: 9

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.

Question 3 - Theory P8

Deciles are denoted by

Options

- A. $D_1, D_2, \dots, D_9$
- B. Q_1, Q_2, Q_3
- C. $P_1, P_2, \dots, P_{99}$
- D. $D_0, D_1, \dots, D_{10}$

Correct answer: $D_1, D_2, \dots, D_9$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $D_1, D_2, \dots, D_9$.

Question 4 - Theory P8

For raw data, the position of D_i is

Options

- A. $(i(n+1))/(10)$
- B. $(i(n+1))/(4)$
- C. $(iN)/(100)$
- D. $((n+i))/(10)$

Correct answer: $(i(n+1))/(10)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i(n+1))/(10)$.

Question 5 - Theory P8

For grouped data, the decile class contains the

Options

- A. $((iN)/(10))^{\text{th}}$ observation
- B. $((iN)/(4))^{\text{th}}$ observation
- C. $((iN)/(100))^{\text{th}}$ observation
- D. N^{th} observation

Correct answer: $((iN)/(10))^{\text{th}}$ observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $((iN)/(10))^{\text{th}}$ observation.

Question 6 - Theory P8

Percentiles divide ordered observations into

Options

- A. 100 equal parts
- B. 10 equal parts
- C. 4 equal parts
- D. 2 equal parts

Correct answer: 100 equal parts

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 100 equal parts.

Question 7 - Theory P8

The number of percentiles is

Options

- A. 99
- B. 100
- C. 9
- D. 3

Correct answer: 99

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 99.

Question 8 - Theory P8

Percentiles are denoted by

Options

- A. $P_1, P_2, \dots, P_{99}$
- B. $D_1, D_2, \dots, D_9$
- C. Q_1, Q_2, Q_3
- D. $P_0, P_1, \dots, P_{100}$

Correct answer: $P_1, P_2, \dots, P_{99}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $P_1, P_2, \dots, P_{99}$.

Question 9 - Theory P8-9

For raw data, the position of P_i is

Options

- A. $(i (n + 1))/(100)$
- B. $(i (n + 1))/(10)$
- C. $(i N)/(4)$
- D. $((n + i))/(100)$

Correct answer: $(i (n + 1))/(100)$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i (n + 1))/(100)$.

Question 10 - Theory P9

For grouped data, the percentile class contains the

Options

- A. $(i N)/(100)$ th observation
- B. $(i N)/(10)$ th observation
- C. $(i N)/(4)$ th observation
- D. N^{th} observation

Correct answer: $(i N)/(100)$ th observation

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $(i N)/(100)$ th observation.

Question 11 - Theory P9

Which relation is correct?

Options

- A. $Q_2 = D_5 = P_{50}$
- B. $Q_1 = D_5 = P_{50}$
- C. $Q_3 = D_5 = P_{50}$
- D. $Q_2 = D_2 = P_{20}$

Correct answer: $Q_2 = D_5 = P_{50}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_2 = D_5 = P_{50}$.

Question 12 - Theory P9

Which relation is correct?

Options

- A. $Q_1 = P_{25}$
- B. $Q_1 = P_{50}$
- C. $Q_2 = P_{25}$
- D. $Q_3 = P_{25}$

Correct answer: $Q_1 = P_{25}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_1 = P_{25}$.

Question 13 - Theory P9

Which relation is correct?

Options

A. $Q_3 = P_{75}$

B. $Q_3 = P_{25}$

C. $Q_2 = P_{75}$

D. $Q_1 = P_{75}$

Correct answer: $Q_3 = P_{75}$

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives $Q_3 = P_{75}$.

Question 14 - Let's Think P9

D_1 is equivalent to

Options

A. P_{10}

B. P_1

C. P_{25}

D. P_{50}

Correct answer: P_{10}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{10} .

Question 15 - Let's Think P9

P_{30} is equivalent to

Options

A. D_3

B. D_1

C. D_5

D. D_7

Correct answer: D_3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives D_3 .

Question 16 - Let's Think P9

D_7 is equivalent to

Options

A. P_{70}

B. P_7

C. P_{35}

D. P_{75}

Correct answer: P_{70}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{70} .

Question 17 - Theory P10

D_4 identifies the value below which approximately what percentage of observations lie?

Options

A. 40%

B. 4%

C. 25%

D. 60%

Correct answer: 40%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40%.

Question 18 - Theory P10

P_{21} identifies the value below which approximately what percentage of observations lie?

Options

A. 21%

B. 79%

C. 2.1%

D. 50%

Correct answer: 21%

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 21%.

Question 19 - Theory P11

For grouped data, N represents

Options

- A. total frequency
- B. class frequency only
- C. class width
- D. lower boundary

Correct answer: total frequency

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives total frequency.

Question 20 - Theory P11

The class containing $(i N)/(10)$ is called the

Options

- A. i^{th} decile class
- B. i^{th} quartile class
- C. median class only
- D. percentile class

Correct answer: i^{th} decile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives i^{th} decile class.

Question 21 - Theory P11

The class containing $(i N)/(100)$ is called the

Options

- A. i^{th} percentile class
- B. i^{th} decile class
- C. quartile class
- D. modal class

Correct answer: i^{th} percentile class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives i^{th} percentile class.

Question 22 - Theory P12

A missing frequency can be found from a known partition value by

Options

- A. substituting it in the appropriate partition formula
- B. using only the range
- C. ignoring cumulative frequency
- D. using the class mark as N

Correct answer: substituting it in the appropriate partition formula

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives substituting it in the appropriate partition formula.

Question 23 - Theory P13

If data are given by mid-values, class boundaries are obtained using

Options

- A. half the difference between consecutive mid-values
- B. the total frequency
- C. the largest frequency
- D. the median only

Correct answer: half the difference between consecutive mid-values

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives half the difference between consecutive mid-values.

Question 24 - Theory P14

The limits of the middle 60% of observations are

Options

- A. P_{20} and P_{80}
- B. P_{30} and P_{70}
- C. P_{40} and P_{60}
- D. P_{10} and P_{90}

Correct answer: P_{20} and P_{80}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{20} and P_{80} .

Question 25 - Theory P14

The limits of the middle 40% of observations are

Options

- A. P_{30} and P_{70}
- B. P_{20} and P_{80}
- C. P_{40} and P_{60}
- D. Q_1 and Q_3

Correct answer: P_{30} and P_{70}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{30} and P_{70} .

Question 26 - Solved Example P9

For $n = 10$, the position of D_3 isth observation

Correct answer: 3.3

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 3.3.

Question 27 - Solved Example P9

For $n = 10$, the position of P_{70} isth observation

Correct answer: 7.7

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 7.7.

Question 28 - Theory P10

For $n = 95$, the position of D_4 isth observation

Correct answer: 38.4

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 38.4.

Question 29 - Theory P10

For $n = 95$, the position of P_{55} isth observation

Correct answer: 52.8

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 52.8.

Question 30 - Solved Example P11

For $n = 15$, the position of D_6 isth observation

Correct answer: 9.6

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 9.6.

Question 31 - Solved Example P11

For $n = 15$, the position of P_{85} isth observation

Correct answer: 13.6

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 13.6.

Question 32 - Solved Example P9

For the ordered data :

169, 225, 289, 324, 325, 400, 625, 729, 784, 841.

Calculate D_3

Correct answer: 299.5

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 299.5.

Question 33 - Solved Example P9

For the ordered data :

169, 225, 289, 324, 325, 400, 625, 729, 784, 841.

Calculate P_{70}

Correct answer: 697.8

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 697.8.

Question 34 - Solved Example P10

No. of defective products	30	35	40	45	50	55	60
No. of firms	12	35	10	15	8	7	8

Calculate D_4 .

Correct answer: 35

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 35.

Question 35 - Solved Example P10

No. of defective products	30	35	40	45	50	55	60
No. of firms	12	35	10	15	8	7	8

Calculate P_{55} .

Correct answer: 40

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 40.

Question 36 - Solved Example P10-11

Profit (in ₹ lakh)	0.5-4.5	5.5-9.5	10.5-14.5	15.5-19.5	20.5-24.5
No. of firms	7	18	25	30	20

Calculate D_4 .

Correct answer: 13

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 13.

Question 37 - Solved Example P10-11

Profit (in ₹ lakh)	0.5-4.5	5.5-9.5	10.5-14.5	15.5-19.5	20.5-24.5
No. of firms	7	18	25	30	20

Calculate P_{21} .

Correct answer: 8.89

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives 8.89.

Question 38 - Theory P14

If 75% of observations lie below a value, that value is

Options

A. Q_3 or P_{75}

B. Q_1 or P_{25}

C. Q_2 or P_{50}

D. D_3 or P_{30}

Correct answer: Q_3 or P_{75}

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives Q_3 or P_{75} .

Question 39 - Theory P14

If 20% of observations lie below a value, that value is

Options

- A. P_{20} or D_2
- B. P_{80} or D_8
- C. Q_1
- D. Q_2

Correct answer: P_{20} or D_2

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives P_{20} or D_2 .

Question 40 - Theory P15

When the desired partition position falls in a class, interpolation uses the frequency of

Options

- A. that partition class
- B. the first class
- C. the last class
- D. the modal class only

Correct answer: that partition class

Explanation and steps

Apply the definition, formula or graphical rule stated in Unit 1. This gives that partition class.

Question 41 - Practice Example P15

Delivery time (minutes)	0-10	10-20	20-30	30-40	40-50
Number of orders	12	18	30	25	15

Find the percentage of orders delivered in less than 25 minutes.

Correct answer: 45%

Explanation and steps

The classes are already continuous. Let $P_i = 25$, where i is the required percentage. The percentile class is 20-30, so $L = 20$, $h = 10$, $f = 30$, c.f. = $12 + 18 = 30$ and $N = 100$. Using $P_i = L + (h/f)(iN/100 - \text{c.f.})$:

$$25 = 20 + (10/30)(i - 30)$$

$$15 = i - 30$$

$$i = 45$$

Therefore, 45% of the orders were delivered in less than 25 minutes.

Question 42 - Practice Example P15

Marks	10-19	20-29	30-39	40-49	50-59
Number of students	8	12	40	25	15

Find the percentage of students who obtained less than 35 marks.

Correct answer: 42%

Explanation and steps

The inclusive classes must first be made continuous: 9.5-19.5, 19.5-29.5, 29.5-39.5, 39.5-49.5 and 49.5-59.5. Let $P_i = 35$, where i is the required percentage. The percentile class is 29.5-39.5, so $L = 29.5$, $h = 10$, $f = 40$, $c.f. = 8 + 12 = 20$ and $N = 100$. Using $P_i = L + (h/f)(iN/100 - c.f.)$:

$$35 = 29.5 + (10/40)(i - 20)$$

$$22 = i - 20$$

$$i = 42$$

Therefore, 42% of the students obtained less than 35 marks.